Voyant Tools Exploratory Analysis Worksheet

This worksheet gives students a structure for working with Voyant Tools and using computational text analysis to consider literature from a new perspective.

Activity in Action: Exploratory Analysis with Voyant

Course: HUM2DH3 Creative, Collaborative, Critical: Introduction to Digital Scholarship (McMaster University, 2023)
Text: *Frankenstein* – Mary Shelley (1818)

In this activity, I created documents that divided *Frankenstein* by speaker/writer and by larger groups like class and gender. This allowed students to compare characters' speech/writing on Voyant.

1. Navigate to Avenue 2 Learn and download the appropriate files under Content -> Voyant Tools Discussion Post Materials. I recommend saving them to your Desktop for easy access.
2. Open Voyant Tools at https://voyant-tools.org/
3. Upload two files of your choice to Voyant. Choose speakers or characters that you find interesting and would like to compare to each other. Click Upload, then navigate to your file location (perhaps your Desktop) and select both files by holding the shift key or by clicking the checkboxes to the left of the file name. (This layout may differ depending on your OS).
4. Wait a few seconds for Voyant to process your texts.

Students formed small groups and worked through the following questions.

Questions to Consider:

1. Look at **Cirrus/Word Cloud.** Which words are the most frequently used in each character's speech/text? Do any words surprise you? Which and why? Do the characters speak like each other or not? Does that align with your idea of the characters?

2. Continue studying **Cirrus/Word Cloud**. Now hone in on specific kinds of words. Think about verbs (action words), adjectives (descriptive words), nouns (people, places, things), and proper nouns (names). When we focus on sub-topics like these, do your findings change or stay the same?

3. Look at the **Document Summary** and examine **Document Length**. Whose text/speech is longer and by how much? What might this suggest about whose perspective dominates the novel? Based on textual data about the quieter character, how might the novel change if they dominated the text?

4. Look at the **Document Summary** and examine **Vocabulary Density.** Remember, a higher number is actually worse here! It means a person uses the same words over and over again. A lower number means a character's vocabulary is more flexible and diverse. Who has a more flexible vocabulary? Does this surprise you or work with your understanding of the character?

5. Look at the **Document Summary** and examine **Average Words per Sentence**. Whose sentences are longer? Whose are shorter? What does this information suggest about each character?

6. Look at **Document Summary** and examine each character's **Distinctive Words**. Do these words have a theme? Do you feel they accurately represent the character?

7. Click on the window icon on the upper right corner. Hover over **Corpus Tools** then select **Collocates** from the menu. Scroll to the bottom of the page and click Scale, then select Documents, then select one character. Examine their Collocates (words that appear close to each other). What kinds of words tend to occur close to each other? Are there any repeated phrases or patterns here? What larger meaning might this pattern suggest? Repeat this exercise with the other character.

8. Look at **Trends**. This tool graphs how often each character uses the corpus's most frequent terms. When you see the distribution of these terms between the characters, does anything stand out? Try adding some other terms from the Word Cloud to the text box and seeing if this presents new information.

After 40 minutes, we shared our findings. Students made insightful discoveries about how computational text analysis emphasizes how gender leads to different experiences. Male characters use active words associated with nature and adventure while women are associated with words that evoke interiors and emotions. Students also compared the language of Victor and the Creature, confirming the strange similarities between these alleged nemeses.